column-hse-ltau: hydrostatic columns on an optical-depth grid

Physics and method of `run.mode = "column-hse-ltau"` in HSEMHS. A self-contained description.

Abstract

A Stokes inversion returns the temperature and the magnetic field of the solar atmosphere on surfaces of constant optical depth, not at geometric heights. This mode turns such an atmosphere into a physical one: for every pixel column it computes the gas pressure and the density on each optical-depth level and the geometric height at which the level lies, assuming that each column is in hydrostatic equilibrium on its own. We derive the hydrostatic equation in optical depth, show why it needs no externally supplied reference pressure and what replaces it, describe the closure at the top of the column, the numerical integration (including the steep pressure rise below hot-over-cool interfaces), and the rule that places the columns relative to each other in height by equalising the total (gas plus magnetic) pressure. All assumptions are listed.

1. Introduction

When we observe the Sun we receive light, and the light from a given depth is attenuated by the gas above it. A spectral line inversion therefore naturally describes the atmosphere as a function of **optical depth** τ_{500} (the optical depth in the continuum at 500 nm): it gives T and $\mathbf{B}$ at, for example, $\log_{10}\tau_{500} = -8.0, -7.9, \ldots, +1.4$ in every pixel. Such an atmosphere has no pressure, no density and no geometry: we do not know how many kilograms of gas a given τ interval contains, nor how many kilometres thick it is. Both follow from two physical laws, hydrostatic equilibrium and the dependence of the opacity on the state of the gas.

Input: the $\log_{10}\tau_{500}$ levels, the horizontal coordinates x, y, and T, B_x, B_y, B_z on every level of every column; the gravity g. **Output:** on the same τ levels, P, ρ and the height z, plus the height of $\tau_{500} = 1$ in every column. T and $\mathbf{B}$ are copied unchanged.

2. Notation

Symbol	Meaning	Unit	Role
$x,\ y$	horizontal coordinates of the pixels, x_i, y_j, uniform spacing Δx, Δy	m	given
τ	optical depth at 500 nm, counted from the top downward	–	coordinate
ℓ_k	$\log_{10}\tau_k$ of level $k = 1 \ldots N_\tau$, increasing (top first), equally spaced ($\Delta\ell = 0.1$)	–	given
s	$\ln\tau = \ell \ln 10$, the integration variable	–	coordinate
$T,\ X$	temperature on the levels; $X = \ln T$	K	given, fixed
$\mathbf{B}$	magnetic field on the levels, (B_x, B_y, B_z)	T	given, fixed
$P,\ u$	gas pressure on the levels; $u = \ln(P/1\,\mathrm{Pa})$	Pa, –	unknown
z	geometric height of a level in its column (upward)	m	unknown
ρ	mass density $\rho(T, P)$	$\mathrm{kg\,m^{-3}}$	derived
χ	extinction coefficient at 500 nm, $\chi_{500}(T, P)$	$\mathrm{m^{-1}}$	derived
κ	extinction per unit mass, $\kappa = \chi/\rho$	$\mathrm{m^2\,kg^{-1}}$	derived
q	ρ/P	$\mathrm{s^2\,m^{-2}}$	derived
m	column mass above a point per unit area	$\mathrm{kg\,m^{-2}}$	derived
H	pressure scale height $P/(\rho g)$	m	derived
P_{tot}	total pressure $P + B^2/(2\mu_0)$	Pa	derived
$P_{\mathrm{tot}}^{\mathrm{ref}}$	reference total pressure used to align the columns	Pa	derived
g	gravitational acceleration, 274	$\mathrm{m\,s^{-2}}$	parameter
μ_0	vacuum permeability $4\pi \times 10^{-7}$	$\mathrm{T\,m\,A^{-1}}$	constant

3. Optical depth measures mass

Going down by a height dz (with z upward, $dz < 0$) the light from below has to cross an additional layer, and the optical depth grows by

$$d\tau = -\chi\, dz. \tag{1}$$

Write $\chi = \kappa\rho$: the extinction is the amount of gas, ρ, times how strongly each kilogram of it removes light, κ. The mass of the column above a point grows by $dm = -\rho\, dz$ over the same step. Therefore

$$d\tau = \kappa\, dm. \tag{2}$$

Optical depth is column mass weighted by the opacity. For gas with a constant κ, $\tau = \kappa m$ exactly. This is the key to the whole mode: an optical-depth scale carries information about how much mass lies above each level.

4. Hydrostatic equilibrium in optical depth

4.1 Derivation

A thin horizontal slab of gas at rest carries its own weight by the pressure difference across it. With z upward this is the hydrostatic equation $dP/dz = -\rho g$ (the vertical component of $-\nabla P - \rho g \hat{\mathbf{z}} = 0$). Since $dm = -\rho\, dz$ it can also be written $dP = g\, dm$: the pressure at a point is the weight of the gas above it. Dividing by Eq. (2):

$$\boxed{\frac{dP}{d\tau} = \frac{g}{\kappa(T,P)}} \tag{3}$$

Adding the optical depth $d\tau$ means adding the mass $d\tau/\kappa$, whose weight is $g\, d\tau/\kappa$. For constant κ, $P = g\tau/\kappa$.

4.2 The form that is integrated

The levels are equally spaced in $\log_{10}\tau$, and τ spans ten decades. We therefore integrate in $s = \ln\tau$ ($d\tau = \tau\, ds$) and in $u = \ln P$ ($dP = P\, du$). With $\kappa = \chi/\rho$:

$$\boxed{\frac{du}{ds} = \frac{g\,\tau}{\kappa\, P} = \frac{g\,\tau\,\rho}{\chi\, P} = \frac{g\,\tau\, q(T,P)}{\chi(T,P)}} \tag{4}$$

and, from Eq. (1), the geometric thickness of the layers:

$$\boxed{\frac{dz}{ds} = -\frac{\tau}{\chi(T,P)}} \tag{5}$$

Where the gas is opaque (large χ) a step in $\ln\tau$ is a thin layer; where it is transparent, the same step is a thick one. Equations (4) and (5) are two coupled ordinary differential equations along each column. ρ and χ come from the equation-of-state table described next.

5. The equation of state and the opacity

Every force balance in this document involves the density ρ, and every optical depth involves the extinction coefficient χ_{500}. Neither is an unknown of its own: at given temperature and pressure both are fixed by the microphysics of the gas. This section explains how.

5.1 The ideal gas

In the solar photosphere and chromosphere the particles interact only weakly, so the gas obeys the ideal-gas law. If n is the number of free particles (atoms, ions, molecules and electrons) per cubic metre,

$$P = n\, k_B T. \tag{6}$$

If the average mass of a free particle is $\bar{\mu}\, m_{\mathrm{u}}$ (with m_{u} the atomic mass unit and $\bar{\mu}$ the **mean molecular weight**), the mass density is $\rho = n\,\bar{\mu}\, m_{\mathrm{u}}$, and therefore

$$\rho = \frac{\bar{\mu}\, m_{\mathrm{u}}}{k_B}\,\frac{P}{T}. \tag{7}$$

At fixed $\bar{\mu}$ and T the density is proportional to the pressure.

5.2 Why $\bar{\mu}$ depends on T and P: ionisation and molecules

For neutral solar gas (about 91 % hydrogen and 9 % helium atoms by number, plus a small fraction of heavier elements) $\bar{\mu} \approx 1.3$. When a hydrogen atom is ionised it becomes two free particles, a proton and an electron, of almost the same total mass; in fully ionised gas $\bar{\mu} \approx 0.6$. In the coolest layers hydrogen partly forms $\mathrm{H_2}$ molecules, which raises $\bar{\mu}$. The degree of ionisation of an element in stage i is given by the **Saha equation**

$$\frac{n_{i+1}\, n_e}{n_i} = \frac{2U_{i+1}(T)}{U_i(T)} \left(\frac{2\pi m_e k_B T}{h^2}\right)^{3/2} e^{-E_i/k_B T}, \tag{8}$$

with E_i the ionisation energy (13.6 eV for hydrogen), U_i the partition functions, m_e the electron mass and h Planck's constant. Two properties matter for us. (i) Because of the exponential, ionisation switches on over a narrow range of temperature. (ii) The electron density n_e appears on the left: at higher density there are more electrons to recombine with, and a higher temperature is needed for the same ionisation. Hence $\bar{\mu} = \bar{\mu}(T,P)$, and Eq. (7) is no longer a simple proportionality. The Saha equations of all elements are coupled through the common n_e and through charge conservation (n_e equals the total charge of the ions), and the molecular equilibria ($\mathrm{H_2}$, $\mathrm{H_2^+}$, $\mathrm{H^-}$) are solved together with them. This assumes **local thermodynamic equilibrium** (LTE): the level populations and ionisation stages are those of a gas in equilibrium at the local T.

5.3 The 500 nm extinction

Light of wavelength 500 nm travelling a distance ds loses the fraction $\chi_{500}\,ds$ of its intensity, by absorption and by scattering. In the solar photosphere χ_{500} is dominated by the negative hydrogen ion H^- (a hydrogen atom with a second, weakly bound electron, binding energy 0.75 eV), whose abundance is proportional to the electron density. The electrons come mostly from easily ionised metals (Na, Mg, Si, Ca, Fe, ...), so χ_{500} grows steeply with temperature and with density. The code adds the continuum contributions of H (bound–free and free–free), H^-, H_2^+, He, He^-, the metals C, N, Mg, Si, Ca and Fe, Rayleigh scattering on H, He and H_2, and Thomson scattering on free electrons, each counted once. Spectral lines are not included: χ_{500} is a continuum extinction.

5.4 The table and its interpolation

Solving the coupled Saha, molecular and charge equations at every grid point at every step would be far too slow. They are therefore solved once, with Wittmann's equation-of-state code for solar composition (with a dedicated hot branch above a switch temperature $T_{\rm sw}$, where the original solver fails in the hot, low-pressure corner), on a grid of $X = \ln T$ and $Y = \ln P$, and stored in a table. For every node the table holds the three functions

$$\ln\rho(X,Y), \qquad \ln n_e(X,Y), \qquad \ln\chi_{500}(X,Y),$$

together with their derivatives ∂_X, ∂_Y and $\partial_X\partial_Y$. Between the nodes each function is evaluated by **bicubic Hermite interpolation**: a cubic polynomial in each direction that matches the stored values and derivatives at the four corners of the cell. The result is continuous with continuous first derivatives, which is important for the Newton-type solvers of the magnetohydrostatic modes, and it provides the derivative $\partial\ln\rho/\partial\ln P|_T$ that they need. The table used by all modes is `cache/eos_table_fullz.h5`. Its range of T and P (its *envelope*) covers the input atmospheres; a point outside the envelope is reported as an equation-of-state failure, never extrapolated.

Two derived quantities appear repeatedly:

$$q \equiv \frac{\rho}{P} \quad [\mathrm{s^2\,m^{-2}}], \qquad H \equiv \frac{P}{\rho\,g} = \frac{1}{g\,q} = \frac{k_B T}{\bar\mu\,m_{\rm u}\,g} \quad [\mathrm{m}]. \tag{9}$$

H is the **pressure scale height**. For $T = 6000$ K and $\bar\mu = 1.3$, $H \approx 140$ km.

6. The boundary conditions

6.1 Why no reference pressure is needed

On a geometric height grid the hydrostatic equation $du/dz = -g\bar\mu m_{\rm u}/(k_B T)$ does not contain the pressure itself when $\bar\mu$ is constant: if $P(z)$ is a solution, so is $c\,P(z)$ for any constant c. A height scale says nothing about how much gas there is, and a reference pressure must be prescribed somewhere. On an optical-depth grid the situation is different. The coordinate τ already counts the opacity-weighted mass (Eq. 2), and Eq. (3) has a natural starting point: at $\tau = 0$ there is no gas above, so $P = 0$. Integrating,

$$P(\tau) = \int_0^\tau \frac{g}{\kappa}\,d\tau'. \tag{10}$$

Once $T(\tau)$ and the opacity law $\kappa(T,P)$ are known, the pressure follows; no value has to be supplied from outside. Equivalently, Eq. (3) is not invariant under $P \to cP$, because κ depends on P (the H^- opacity grows with the electron density).

6.2 What replaces it: the top closure

The grid does not start at $\tau = 0$ but at $\tau_1 = 10^{-8}$. Above the top level we assume that κ keeps its value κ_1 and that $P(\tau = 0) = 0$. Equation (10) then gives

$$P_1 = \frac{g\,\tau_1}{\kappa(T_1,P_1)} = \frac{g\,\tau_1\,\rho(T_1,P_1)}{\chi(T_1,P_1)}. \tag{11}$$

P_1 appears on both sides, because κ depends on P. It is found by a damped fixed-point iteration in u: start from $u = \ln(100)$, compute $u_{\rm new} = \ln(g\tau_1\rho/\chi)$ at the current u, set $u \leftarrow \frac{1}{2}(u + u_{\rm new})$, and repeat until the change is below 10^{-10} (at most 200 times). The damping avoids oscillation.

How much does this assumption matter? At depth τ the pressure is P_1 plus the weight of the gas between τ_1 and τ. If κ were constant, $P_1/P(\tau) = \tau_1/\tau$, i.e. 10^{-8} at $\tau = 1$. Any reasonable error in κ above the top therefore affects only the uppermost few levels. The result is free of an external pressure, but it depends on modelling choices instead: this top closure, the interpolation of the temperature between levels (which matters most at sharp temperature jumps, Sect. 8.3) and, for the heights, the alignment of the columns (Sect. 9). It is also only as good as the opacity: if the τ scale of the input was computed with a different opacity, the pressures differ in proportion.

6.3 The height: a constant of integration per column

Equation (5) fixes the *distances* between the levels of a column, not where the column sits. Each column is first integrated with $z = 0$ at the top level and then shifted so that $z = 0$ at $\log_{10}\tau = 0$. Its absolute height is fixed afterwards by the alignment rule of Sect. 9. So the reference that this mode needs is a reference *height*, not a reference pressure.

7. Assumptions

A1. Static gas, vertical balance only. Each column is hydrostatic on its own; velocities and the horizontal force balance between columns are ignored.
A2. No Lorentz force in the column balance. The field enters only the alignment, through its magnetic pressure; its tension is ignored.
A3. Consistent opacity. The input τ_{500} scale is assumed to be the continuum optical depth at 500 nm along the vertical computed with the same LTE equation of state and opacity as the table.
A4. Top closure: κ constant above the top level and $P(\tau = 0) = 0$ (Eq. 11).
A5. Temperature between levels: $\ln T$ varies linearly with $\ln\tau$ inside each interval. Across a sharp temperature jump this choice determines how much mass the interval holds (Sect. 8.3).
A6. Alignment: at one common height all columns have the same total pressure $P + B^2/2\mu_0$; below the deepest level the total pressure is extrapolated with the gas-pressure scale height (Sect. 9).
A7. Gauge: the mean height of $\tau_{500} = 1$ is zero.
A8. Constant, vertical gravity and plane-parallel geometry.
A9. Field attached to the τ levels: the given field values stay on their levels wherever those end up in space; nothing makes them divergence-free there.

8. Numerical integration of a column

8.1 Temperature inside an interval

Between levels k and $k+1$ ($s_k = \ell_k \ln 10$, $s_{k+1} = \ell_{k+1}\ln 10$),

$$X(s) = X_k + (X_{k+1} - X_k)\,\frac{s - s_k}{s_{k+1} - s_k}. \tag{12}$$

8.2 Heun's method with adaptive substeps

Write Eqs. (4)–(5) as $du/ds = f(s,u)$, $dz/ds = h(s,u)$. The integration runs downward, from s_k to s_{k+1}, in substeps of length δ. One substep from (s,u,z) is Heun's predictor–corrector method (the trapezoidal rule with an Euler predictor):

$$\begin{aligned} &f_1 = f(s,u), \quad h_1 = h(s,u), \qquad \tilde{u} = u + \delta f_1 \ \text{ (predictor)}, \qquad f_2 = f(s+\delta,\tilde{u}), \quad h_2 = h(s+\delta,\tilde{u}),\\ &u \leftarrow u + \tfrac{\delta}{2}(f_1 + f_2), \qquad z \leftarrow z + \tfrac{\delta}{2}(h_1 + h_2), \qquad s \leftarrow s + \delta. \end{aligned} \tag{13}$$

The method is second-order accurate: the error falls with the square of the substep length. The substep length δ is chosen so that the slope is resolved at *both* ends of the substep:

1. start from $\delta = \min\left(s_{k+1} - s,\ 0.05/|f_1|\right)$, so that $\ln P$ would change by at most 0.05 (a 5 % change in pressure) at the slope of the start;
2. evaluate f_2 at the end; while $\delta\,|f_2| > 0.1$, divide δ by 4 and evaluate again (down to 10^{-9} of the interval).

The second condition matters where the slope grows rapidly within an interval (next subsection): a step sized from f_1 alone would average a slope that is a thousand times larger at its end over a long step and overshoot the pressure by orders of magnitude. Intervals with a steep pressure rise therefore receive many short substeps.

8.3 Hot gas over cool gas

The 500 nm opacity rises very steeply with temperature (Sect. 5), so cool gas is far more transparent than hot gas. Suppose that within one grid interval a hot layer (8000 K) lies directly on a cool one (4000 K). With the linear interpolation of $\ln T$, part of that tenth of a decade in τ is filled with nearly transparent gas. With $P = g\,m$ (the pressure is the weight of the column mass m above), Eq. (3) reads

$$\frac{d\ln m}{d\ln\tau} = \frac{\tau}{\kappa\, m}\,: \tag{14}$$

collecting optical depth from transparent gas (small κ) genuinely requires a large column mass, and the pressure below a hot-over-cool interface rises steeply. The mode does not limit this rise: it is what hydrostatic equilibrium predicts for the given $T(\tau)$, and the step rule above resolves it. How large the rise comes out does depend on how the temperature is interpolated between the two levels, since the true structure between them is unknown.

The heights stay well behaved: since $dz = -dm/\rho$ with $dm > 0$ going down, the heights strictly decrease with depth, however steep the pressure rise.

8.4 Failure of a column

If the pressure leaves the equation-of-state table (it would exceed the upper edge of the table minus 1 in $\ln P$) or a slope becomes non-finite, the column is *not* altered to make it fit: its pressures from that level down and all its heights are marked undefined, and the column is counted as failed. If any column fails, the run stops with an equation-of-state error, before the alignment (which would otherwise spread the undefined values to all columns).

8.5 The result of a column

After the last interval each column has u_k and z_k on all levels, with $z_1 = 0$. The heights are shifted so that $z = 0$ at $\ell = 0$, the value at $\ell = 0$ being interpolated linearly in ℓ (or extrapolated from the end interval).

9. Where the columns sit: total-pressure alignment

9.1 Magnetic pressure

After Sect. 8 every column has a correct internal structure but an arbitrary vertical position. To place the columns relative to each other we use the horizontal force balance in its simplest form. The magnetic (Lorentz) force can be rewritten with a vector identity:

$$\mathbf{J} \times \mathbf{B} = \frac{(\nabla \times \mathbf{B}) \times \mathbf{B}}{\mu_0} = -\nabla\left(\frac{B^2}{2\mu_0}\right) + \frac{(\mathbf{B} \cdot \nabla)\mathbf{B}}{\mu_0}. \tag{15}$$

The first term acts like an additional pressure, the **magnetic pressure** $B^2/2\mu_0$; the second is a tension along curved field lines.

9.2 The alignment rule

If the field is nearly vertical and weakly curved, its tension is small, and the horizontal force balance between neighbouring columns at the same height reduces to equal total pressure:

$$\frac{\partial}{\partial x}\left(P + \frac{B^2}{2\mu_0}\right) = \frac{\partial}{\partial y}\left(P + \frac{B^2}{2\mu_0}\right) = 0 \qquad \text{(tension neglected)}. \tag{16}$$

A column with a higher total pressure would push its neighbour aside. The mode imposes this at *one* height only:

1. In every column compute $\ln P_{\mathrm{tot,k}} = \ln\bigl(P_k + B_k^2/2\mu_0\bigr)$ on all levels.
2. The reference is the median over all columns of the total pressure on the deepest level, $P_{\mathrm{tot}}^{\mathrm{ref}} = \mathrm{median}_{i,j}\, P_{\mathrm{tot},N_\tau}$.
3. In each column find the height z_r at which $P_{\mathrm{tot}} = P_{\mathrm{tot}}^{\mathrm{ref}}$: go up from the deepest level to the first interval in which $\ln P_{\mathrm{tot}}$ drops below $\ln P_{\mathrm{tot}}^{\mathrm{ref}}$, and interpolate linearly in z between its two levels.
4. For a column whose deepest total pressure is already below the reference, the crossing lies below the grid. It is extrapolated with the gas-pressure scale height of the deepest level, $z_r = z_{N_\tau} - \bigl(\ln P_{\mathrm{tot}}^{\mathrm{ref}} - \ln P_{\mathrm{tot},N_\tau}\bigr)\, H_{N_\tau}$ with $H = P/(\rho g)$. This is an approximation (the total pressure does not fall with exactly the gas scale height); the number of such columns is reported.
5. Shift the column: $z_k \leftarrow z_k - z_r$. Now $P_{\mathrm{tot}} = P_{\mathrm{tot}}^{\mathrm{ref}}$ at $z = 0$ in every column.

With `align = "none"` this step is skipped and every column keeps $z = 0$ at $\tau_{500} = 1$.

9.3 Consequence: the Wilson depression

Under this rule, a column with a strong field carries part of the total pressure as magnetic pressure, so it needs less gas pressure at the reference height. Less gas pressure means lower density and lower opacity, so light escapes from deeper: its τ surfaces lie lower. This is the **Wilson depression** seen in sunspots observed near the limb. It is an expected tendency, not a guarantee: the outcome for a given column also depends on its temperature structure and on the approximations above.

9.4 The gauge

Finally all columns are shifted together by the same amount so that the mean over all columns of the height of $\tau_{500} = 1$ is zero. This is only a choice of origin.

10. Iteration and acceptance criteria

The mode contains no global iteration. Its convergence criteria are local:

- the top pressure: change of u below 10^{-10} in the fixed-point iteration of Eq. (11);

- the substep length: $\ln P$ changes by at most about 0.05 at the slope of the start and 0.1 at the slope of the end of every Heun substep;
- every pressure stays inside the equation-of-state table; otherwise the column fails and the run stops (Sect. 8).

Every column is independent until the alignment, which needs one global number (the median reference total pressure) and the gauge, which needs one global mean. Finally the returned fields are checked: all values finite, P, ρ and χ positive, every evaluated state inside the table, and in every column the heights strictly decreasing with depth. The first failed check, if any, is reported.

11. Three dimensions: what the result does and does not satisfy

- **Vertical:** every column obeys $dP/d\tau = g/\kappa$ (hydrostatic equilibrium) to the accuracy of the integration.
- **Horizontal:** the total pressure of all columns is equal at $z = 0$ only. At other heights neighbouring columns generally have different total pressures, so a horizontal force remains; the field's tension is not included at all. This mode is therefore not a three-dimensional equilibrium. Imposing horizontal force balance at all heights, together with the full Lorentz force, requires moving the τ surfaces in height and adjusting the pressure in a coupled way, which is what the mode `mhs-poisson-ltau` does starting from this result.
- **The field:** the given $\mathbf{B}$ values stay on their τ levels. Because the levels move to new heights, the field as it ends up in space need not satisfy $\nabla \cdot \mathbf{B} = 0$; this mode does not use derivatives of $\mathbf{B}$ and is not affected, but any later use of the field should keep it in mind.

12. Summary of the algorithm

1. Read ℓ_k, x, y, T, $\mathbf{B}$; load the equation-of-state table.
2. For every column independently: find $P_1 = g\tau_1\rho/\chi$ by damped fixed-point iteration; integrate $du/ds = g\tau q/\chi$ and $dz/ds = -\tau/\chi$ downward with Heun substeps sized by the slope at both ends, $\ln T$ linear in $\ln \tau$; a column whose pressure leaves the table fails the run; shift so that $z = 0$ at $\tau = 1$.
3. Compute the median total pressure on the deepest level; shift every column so that its total pressure equals it at $z = 0$ (extrapolating below the grid where needed).
4. Shift all columns so that the mean height of $\tau_{500} = 1$ is zero.
5. Evaluate ρ from the table; write z, P, ρ, T, $\mathbf{B}$ and $z(\tau_{500} = 1)$.